RoomCleaner engineering calculations

Expected values behind the presentation charts. Every result is a formula over the inputs on its tab.

Colour key	
Blue text	Input: change it and every result on the tab updates
Green text	Link to another tab
Black text	Formula
Tab	**What it shows**
Cable tension	Tension per cable vs depth below the anchors, against the winch design force
Winch pull	Driver current, running torque and pull at the empty and full spool
Heat fin	Fin efficiency and heat carried vs exposed length, 0.019 in sheet vs 1/8 in plate
Motor temperature	Motor case temperature for open air, vented and sealed shells, with and without the fin
Wiring	Conductor length for one central box vs a controller in every station; voltage drop
Bead homing	Share of the room still reachable when every bead homes at the room centre
Floor board	Position error per frame vs how many tags the camera sees
Lift budget	Lift capacity vs the claw, battery and jeans
Gears	Lewis tooth-bending safety factor at servo stall
References	

Incropera, DeWitt, Bergman, Lavine: Fundamentals of Heat and Mass Transfer, ch. 3 (extended surfaces). Fin efficiency and fin resistance.

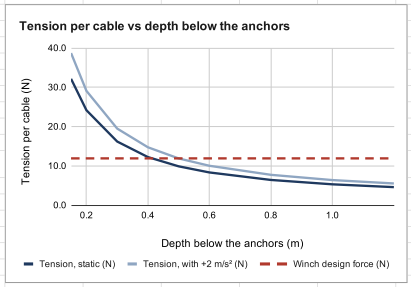
Allegro MicroSystems A4988 datasheet: current limit I_trip = V_ref / (8 R_s).

Cable tension vs depth below the anchors

Centred claw, four equal shares: $T = W / (4 \sin \varphi)$, $\sin \varphi = d / \sqrt{(d^2 + D_h^2)}$. Blue cells are inputs.

Inputs	Value	Unit	Source
Claw mass	0.45	kg	Claw mass budget (requirement)
Test payload	0.454	kg	1 lb first test load
Gravity g	9.81	m/s²	
Design acceleration a	2	m/s²	Requirement
Horizontal distance claw to anchor D_h	2.17	m	Room centre to a corner, 11 × 9 ft room
Winch design force	11.90	N	Winch pull tab
Weight W	8.87	N	(claw + payload) × g

Depth below anchors (m)	sin φ	Tension, static (N)	Tension, with +2 m/s² (N)	Winch design force (N)
0.15	0.069	32.2	38.7	11.9
0.2	0.092	24.2	29.1	11.9
0.3	0.137	16.2	19.5	11.9
0.4	0.181	12.2	14.7	11.9
0.5	0.225	9.9	11.9	11.9
0.6	0.266	8.3	10.0	11.9
0.8	0.346	6.4	7.7	11.9
1	0.419	5.3	6.4	11.9
1.2	0.484	4.6	5.5	11.9



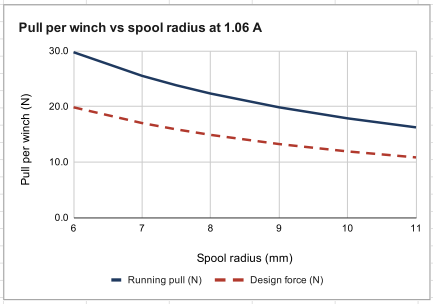
Winch pull from motor torque

Pull = (running / holding ratio) × holding torque × (I / I_rated) / r. Blue cells are inputs.

Inputs	Value	Unit	Source
Holding torque, rated	0.36	N·m	Motor rating at 1.5 A (nameplate to confirm)
Rated current	1.5	A	Motor rating
Vref, measured maximum	0.85	V	Measured on the A4988 carrier, Oct 21
Sense resistor R_s	0.1	Ω	R100 carrier
Running / holding torque ratio	0.7	-	Assumption; the pull test measures it
Skip margin	1.5	-	Design margin against missed steps
Spool radius, empty (core)	7.5	mm	Spool v3, Ø15 core
Spool radius, full	10	mm	Spool v3 with line wound on

Results	Value	Unit	How
Driver current I	1.06 [1]	A	I = Vref / (8 R_s), Allegro A4988 datasheet
Running torque	0.18	N·m	ratio × holding × I / I_rated
Running pull, empty spool	23.80	N	T = τ / r
Running pull, full spool	17.85	N	T = τ / r
Winch design force	11.90	N	full-spool pull / margin

Spool radius (mm)	Running pull (N)	Design force (N)
6	29.8	19.8
7	25.5	17.0
7.5	23.8	15.9
8	22.3	14.9
9	19.8	13.2
10	17.9	11.9
11	16.2	10.8

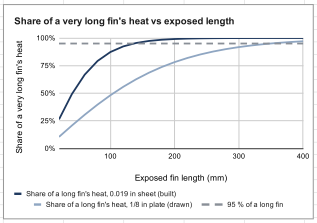


Heat fin: how much length is useful

Straight fin, both faces in room air, insulated tip: m = √(2h / (k t)), η = tanh(mL) / (mL), G = η h (2 w L), R_fin = 1 / G. Source: Incropera, Fundamentals of Heat and Mass Transfer, ch. 3 (extended surfaces). Blue cells are inputs.

Inputs	Value	Unit	Source
Aluminium conductivity k	160	W/m·K	Common sheet alloys 138 to 167
Film coefficient h	7	W/m²·K	Still air plus radiation from a black fin, range 5 to 9 (assumption)
Fin width w	44	mm	Motor width (drawing)
Thickness, as built	0.483	mm	0.019 in sheet (measured)
Thickness, as drawn	3.175	mm	1/8 in plate (drawing)
Exposed length, as built	200	mm	Above the cover's fin pocket

Results	Value	Unit	How
m, as built	13.5	1/m	m = √(2h / (k t))
m, as drawn	5.2	1/m	m = √(2h / (k t))
Length for 95 % of a long fin's heat, as built	136.1	mm	tanh(mL) = 0.95
Length for 95 % of a long fin's heat, as drawn	348.9	mm	tanh(mL) = 0.95
Fin efficiency at the built length, as built	37%	-	η = tanh(mL) / (mL)
Fin resistance at the built length, as built	22.1	K/W	R_fin = 1 / (η h 2 w L)
Fin resistance at the built length, as drawn	10.9	K/W	R_fin = 1 / (η h 2 w L)



Exposed length L (mm)	Efficiency, 0.019 in	Heat per kelvin, 0.019 in (W/K)	Efficiency, 1/8 in	Heat per kelvin, 1/8 in (W/K)	Share of a long fin's heat, 0.019 in sheet (built)	Share of a long fin's heat, 1/8 in plate (drawn)	95 % of a long fin
20	98%	0.012	100%	0.012	26%	10%	95%
40	91%	0.023	99%	0.024	49%	21%	95%
60	83%	0.031	97%	0.036	67%	30%	95%
80	74%	0.036	95%	0.047	79%	40%	95%
100	65%	0.040	92%	0.057	87%	48%	95%
120	57%	0.042	89%	0.065	92%	56%	95%
140	51%	0.044	85%	0.073	95%	63%	95%
160	45%	0.045	82%	0.080	97%	69%	95%
180	41%	0.045	78%	0.087	98%	74%	95%
200	37%	0.045	74%	0.092	99%	78%	95%
220	34%	0.046	71%	0.096	99%	82%	95%
240	31%	0.046	68%	0.100	100%	85%	95%
260	29%	0.046	64%	0.103	100%	88%	95%
280	27%	0.046	61%	0.106	100%	90%	95%
300	25%	0.046	58%	0.108	100%	92%	95%
320	23%	0.046	56%	0.109	100%	93%	95%
340	22%	0.046	53%	0.111	100%	95%	95%
360	21%	0.046	51%	0.112	100%	96%	95%
380	20%	0.046	48%	0.113	100%	96%	95%
400	19%	0.046	46%	0.114	100%	97%	95%

Motor case temperature: shell and fin

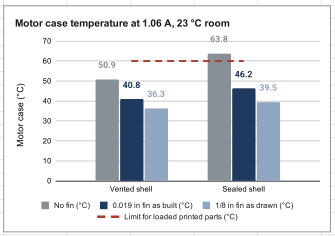
Two-node network: motor case to shell air to room, with the fin as a second path from the motor straight to the room. Steady state. Blue cells are inputs.

Inputs	Value	Unit	Source
Coil resistance per phase	1.8	Ω	Assumption; to be measured with a meter
Driver current	1.06	A	Winch pull tab
Motor heat P_m = I² R	2.03	W	Joule heating
Electronics heat inside the shell	1.5	W	Earlier thermal model (estimate)
Room air near the ceiling	23	°C	Measured 20.6 °C at the floor; +2.4 °C near the ceiling assumed
Contact resistance motor to fin	1	K/W	Assumption
Fin path, 0.019 in (fin + contact)	23.1	K/W	Heat fin tab
Fin path, 1/8 in (fin + contact)	11.9	K/W	Heat fin tab

Enclosure	Motor to shell air (K/W)	Shell air to room (K/W)	Electronics heat (W)	No fin (°C)	Shell heat flow, 0.019 in fin (W)	0.019 in fin as built (°C)	Shell heat flow, 1/8 in fin (W)	1/8 in fin as drawn (°C)
Open air	12	0	0	47.4	1.34	39.0	1.01	35.1
Vented shell	12	1	1.5	50.9	1.26	40.8	0.91	36.3
Sealed shell	14	3.5	1.5	63.8	1.03	46.2	0.64	39.5

Chart data	No fin (°C)	0.019 in fin as built (°C)	1/8 in fin as drawn (°C)	Limit for loaded printed parts (°C)
Vented shell	50.9	40.8	36.3	60
Sealed shell	63.8	46.2	39.5	60

Limit for loaded printed parts	60	°C	Design limit; PETG softens near 80 °C



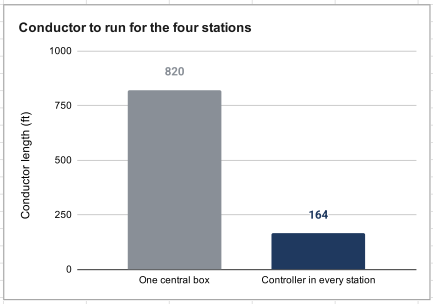
Wiring: one central box vs a controller in every station

Conductor length = conductors per corner × total route length; voltage drop on the longest run V = I · R′ · 2L. Blue cells are inputs.

Inputs	Value	Unit	Source
Route to the station above the supply	6	ft	Room layout, Oct 14
Route along the 9 ft wall	21	ft	Room layout, Oct 14
Route along the 11 ft wall	23	ft	Room layout, Oct 14
Route to the diagonal corner	32	ft	Room layout, Oct 14
Conductors per corner, central box	10	-	Motor 4 + homing switch 2 + encoder 4
Conductors per corner, controller in the station	2	-	12 V and ground
Current per branch (illustrative)	2	A	Illustrative, not measured
Copper resistance, 16 AWG	4.016	Ω per 1000 ft	Standard AWG table
Copper resistance, 18 AWG	6.385	Ω per 1000 ft	Standard AWG table

Results	Value	Unit	How
Total route length	82	ft	sum of the four routes
Conductor, central box	820	ft	conductors × route
Conductor, controller in every station	164	ft	conductors × route
Drop on the longest run, 16 AWG	0.51	V	V = I R′ 2L
Drop on the longest run, 18 AWG	0.82	V	V = I R′ 2L

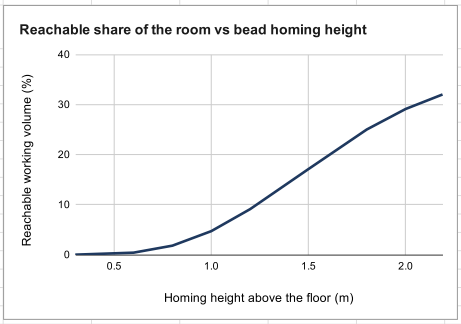
Architecture	Conductor length (ft)
One central box	820
Controller in every station	164



Bead homing: how much of the room stays reachable

A bead cannot pass its outlet, so no cable can get shorter than its length at the homing pose. All four beads homed with the claw at the room centre; working volume gridded with 0.2 m wall and 0.4 m ceiling margins.

Homing height above the floor (m) [2]	Cable length at the homing pose (m)	Reachable share of the working volume (%)
0.3	3.107	0
0.6	2.892	0.4
0.8	2.758	1.8
1	2.633	4.7
1.2	2.517	9.1
1.5	2.364	17.1
1.8	2.241	25
2	2.178	29.1
2.19	2.134	32



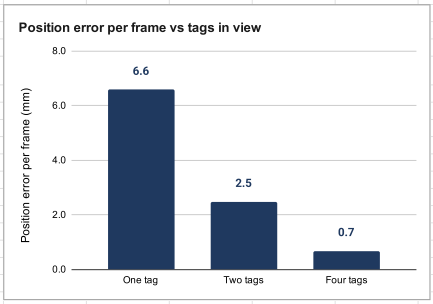
Check: cable length at the homing pose from the geometry		
Anchor height (m)	2.5932	Room 11 × 9 × 9 ft, anchors 0.05 m in from the corners
Anchor to room centre, horizontal (m)	2.096	Half the room diagonal, less the 0.05 m inset
Cable length at 1.0 m homing height (m)	2.633	Matches row 8

Floor board: angle error becomes position error

Lever arm: $\delta x = \sigma_\theta \cdot L$, with L the camera height above the board. Blue cells are inputs.

Inputs	Value	Unit	Source
Camera height above the board L	0.51	m	Claw resting pose

Tags in view	Angle noise per frame (mrad)	Position error per frame (mm)	Source of the angle noise
One tag	12.9	6.6	Corner-noise model, 0.3 px, 520 px focal length
Two tags	4.85	2.5	Same model
Four tags	1.32	0.7	Same model



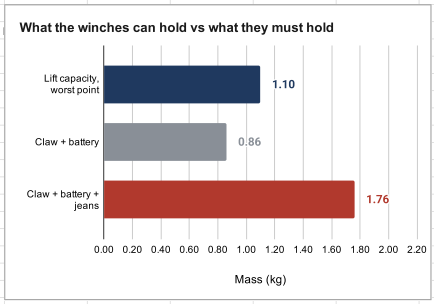
Lift budget: what the winches can hold vs what they must hold

Capacity is the worst point in the working volume at the driver's current limit. The claw has not been weighed; its mass is estimated from the CAD volumes. Blue cells are inputs.

Inputs	Value	Unit	Source
Lift capacity, worst point in the room	1.1	kg	Linear program over the workspace, 0.2 m wall and 0.4 m ceiling margins, 11.9
Printed parts, solid PETG mass in CAD	927	g	Sum of the final gripper's printed part volumes × 1.27 g/cm³
Print fill factor (3 walls, 20 % infill)	0.55	-	Assumption
Aluminium tube	198	g	CAD volume × 2.70 g/cm³
Servo MG996R	55	g	Datasheet
Battery pack	99	g	Tenergy 7.4 V 2200 mAh pack spec
Jeans	0.9	kg	Working load (requirement)

Results	Value	Unit	How
Claw, as printed (estimate)	0.76	kg	printed parts × fill + tube + servo
Claw + battery	0.86	kg	
Claw + battery + jeans	1.76	kg	
Shortfall with jeans	0.66	kg	what has to come from a lighter claw or more pull

Case	Mass (kg)
Lift capacity, worst point	1.10
Claw + battery	0.86
Claw + battery + jeans	1.76



Gear tooth bending at servo stall (Lewis)

σ = F_t / (b m Y), F_t = 2τ / d, d = m z, SF = σ_allow / σ. Blue cells are inputs.

Inputs	Value	Unit	Source
Servo stall torque at 6.0 V	1.079	N·m	MG996R rating, 11.0 kg·cm
Module m	1.5	mm	Gear design
Mesh efficiency (printed)	0.95	-	Assumption
Allowable stress, PETG	22.5	MPa	Half of 45 MPa tensile strength (layer adhesion, fatigue)

Gear	Teeth z	Face width b (mm)	Form factor Y	Torque on the gear (N·m)	Pitch diameter d (mm)	Tangential force F_t (N)	Root stress (MPa)	Safety factor
Pinion (on the servo)	18	15.7	0.309	1.079	27	79.9	10.98	2.05
36-tooth gear (same mesh)	36	15.7	0.3775	2.158	54	79.9	8.99	2.50
24-tooth sync gear, one jaw takes all	24	20	0.337	2.050	36	113.9	11.27	2.00

[1] Source: Allegro MicroSystems A4988 datasheet, current limit I_trip = V_ref / (8 R_s).

[2] Source: calculations/bead_reach.py in the RoomCleaner repository, run Oct 21, 2026 (grid of 60 x 50 x 50 points over the working volume).